CSE 107 — Probability and Statistics for Engineers

Fall 2026 · Section 01 · Course syllabus

This course develops probability models, random variables, statistical reasoning, and elementary stochastic processes. We emphasize clear definitions, justified methods, and the interpretation of calculations, with examples from computing and engineering.

Instructor and course information

Item	Information
Instructor	Jose Antonio Aguirre Perez de Leon Please call me Antonio or Tony. PhD candidate in Statistics, UC Santa Cruz.
Contact	jaguir26@ucsc.edu · antonio-de-leon.com
Course	CSE 107-01 · Class 11896 · 5 units
Lectures	Monday, Wednesday, Friday, 1:20–2:25 p.m. Classroom Unit 002 · In person
First meeting	Sep 25, 2026
No lecture	Nov 11, 2026 and Nov 27, 2026
Student / office hours	All times Pacific. Instructor: Fridays, 3–4 p.m., in person at Graduate Student Commons (GSC), Room 204; additional hours by appointment via email. Jeannine Romero: Tuesday and Thursday, 10–11 a.m. (Zoom). Ehsan Mirafzali: Monday, 3–5 p.m. (Zoom). Seokjun Choi: Wednesday, 3–4 p.m. and 5:05–6:05 p.m. (3–4 Zoom; 5:05–6:05 Zoom).
Teaching assistants	Jeannine Romero · jromer37@ucsc.edu Ehsan Mirafzali · smirafza@ucsc.edu Seokjun Choi · schoi90@ucsc.edu
Optional discussions	Optional, ungraded TA-led practice; no required attendance or submission. All times Pacific. Jeannine Romero: 01A · Friday, 9:20–10:25 a.m.; in person, R Carson Acad 240. Ehsan Mirafzali: 01B · Friday, 10:40–11:45 a.m.; Zoom. Seokjun Choi: 01C · Wednesday, 4–5:05 p.m.; Zoom. Seokjun's regular sessions begin Sep 30, 2026. Sections 01B and 01C meet on Zoom; listed campus rooms do not apply. Check Ed Discussion for meeting exceptions.

Communication and course website

Use Canvas for the course schedule, materials, quizzes, and gradebook. Check Ed Discussion regularly for announcements; course announcements will not be posted in Canvas Announcements. Open CSE 107 from your UCSC Canvas dashboard. Use Ed Discussion in the Canvas course navigation for course questions. Use private messages on Ed Discussion for private, administrative, or accommodation questions. Please do not use Canvas Inbox.

Office hours are a place to ask questions, review a method, or discuss study strategies. Bring an attempted problem or a specific question when possible. Keep diagnoses and other sensitive information out of public discussion boards.

Class format

Lectures combine definitions, derivations, worked examples, and short opportunities to practice. The course meets three times per week. Lecture attendance is not recorded, but brief graded activities take place in class without advance announcement of their dates or times.

Discussion sections are optional. After a brief recap, the TA guides the exercises in order, pausing for student attempts, comparison of methods, and explanation of reasoning. You may also use these exercises for independent study. Missing discussion has no grade consequence and requires no absence report or makeup.

Learning outcomes

- Describe events and sample spaces; count outcomes; apply conditioning, independence, total probability, and Bayes' theorem.
- Use probability mass functions, densities, and cumulative distribution functions for discrete and continuous random variables.
- Find marginal and conditional distributions; calculate expectations, variances, covariance, and elementary distributions of transformed variables.
- Distinguish exact probabilities, bounds, and approximations; apply Markov's and Chebyshev's inequalities, the finite-variance weak law of large numbers, and the central limit theorem under their stated assumptions.
- Interpret sampling variability, standard errors, and an introductory confidence interval for a mean.
- Model Poisson counts and increments. Interpret one-step and two-step transitions of a finite Markov chain; this final introduction is excluded from homework, quizzes, and the final examination.

Prerequisites and preparation

The current course prerequisites are CSE 16 and one of AM 30, MATH 22, or MATH 23A. We use algebra, logarithms, sums, derivatives, and basic integration. Necessary iterated integrals and small matrices are introduced when needed. The Mathematics Review is an ungraded warm-up, not an additional enrollment requirement. Programming is not required.

Reading materials and technology

Course slides, explanations, and posted notes are the primary route. Mor Harchol-Balter's Introduction to Probability for Computing and Morris H. DeGroot and Mark J. Schervish's Probability and Statistics provide complementary explanations and examples. Use either book or consult both; neither whole chapters nor both books are required. Follow the topic selections in the Weekly Guide and weekly Canvas pages. Printed page numbers refer to the book, not the PDF viewer. Optional reading and practice do not expand assessment scope. No textbook purchase is required. Harchol-Balter book access · DeGroot–Schervish PDF.

You need reliable access to Canvas, Gradescope, Ed Discussion, and posted Google Drive documents; a computer is recommended for quizzes and uploads. Homework requires paper

and a scanner or scanning app, or a tablet for stylus handwriting. A basic nonprogrammable, nongraphing calculator is permitted on exams; it is not required for Midterm 1. Library laptop loans and ITS support are listed under Student services.

Weekly routine

Before class, review the posted slides and mark a question. During class, attempt the examples. Afterwards, reconstruct a solution without looking, review definitions, and work on homework in several sessions. Keep your own notes, check deadlines, and ask questions before they accumulate. Optional discussion and office hours can support this work.

Course calendar — opening week through Week 5

All dates below are in 2026 and all times are Pacific. This calendar follows the existing course plan; announced changes will be reflected in Canvas and the course documents. In-class activity dates are not announced in advance. The detailed lesson sequence, reading selections, and resource links are in the Weekly Guide: CSE 107 — Weekly Guide.

Week / dates	Lecture topics and meetings	Assessment milestones
0 Sep 25, 2026	Course introduction; outcomes, events, and finite probability models.	Read the syllabus; begin the ungraded Mathematics Review. No graded Week 0 activity.
1 Sep 28, 2026 – Oct 2, 2026	Events and counting; conditional probability; total probability and Bayes' theorem.	HW1 released. Quiz 1 closes Oct 4, 2026.
2 Oct 5, 2026 – Oct 9, 2026	Independence; random variables and indicators; PMFs and CDFs; Bernoulli and binomial models.	HW1 due Oct 6, 2026. HW2 released. Quiz 2 closes Oct 11, 2026.
3 Oct 12, 2026 – Oct 16, 2026	Expectation and linearity; Midterm 1 review; Midterm 1 on Oct 16, 2026. The expectation lesson is outside Midterm 1 scope.	HW2 due Oct 12, 2026. Quiz 3 closes Oct 14, 2026.
4 Oct 19, 2026 – Oct 23, 2026	Variance and rescaling; joint discrete distributions, marginals, conditional laws, and independence.	HW3 released. Quiz 4 closes Oct 25, 2026.
5 Oct 26, 2026 – Oct 30, 2026	Densities, CDFs, and continuous expectation; exponential and normal models; joint densities and support.	HW3 due Oct 27, 2026. HW4 released. Quiz 5 closes Nov 1, 2026.

Use the weekly lecture slides and notes first. The Weekly Guide lists complementary selections from both textbooks; use either book or both as helpful. Discussion practice is optional. Homework due times and final grace cutoffs are listed below.

Course calendar — Weeks 6–10 and final

Week / dates	Lecture topics and meetings	Assessment milestones
6 Nov 2, 2026 – Nov 6, 2026	Conditional densities and basic conditional means; covariance and variance of sums; elementary CDF transformations.	HW4 due Nov 3, 2026. HW5 released. Quiz 6 closes Nov 8, 2026.
7 Nov 9, 2026 – Nov 13, 2026	Markov and Chebyshev bounds; finite-variance weak LLN; Midterm 2 review. No lecture on Nov 11, 2026.	HW5 due and Quiz 7 closes Nov 12, 2026.
8 Nov 16, 2026 – Nov 20, 2026	Midterm 2 on Nov 16, 2026; CLT for sums and averages; normal approximations.	HW6 released. Quiz 8 closes Nov 22, 2026.
9 Nov 23, 2026 – Nov 27, 2026	Estimation, standard error, and an introductory confidence interval; Poisson distribution and processes. No lecture on Nov 27, 2026.	HW6 due Nov 24, 2026. HW7 released. Quiz 9 opens Nov 25, 2026 and closes Dec 1, 2026.
10 Nov 30, 2026 – Dec 4, 2026	Poisson-process calculations; a brief Markov-chain introduction; cumulative review. Markov chains are excluded from the final.	HW7 due Dec 2, 2026. Quiz 10 closes Dec 4, 2026.
Finals Dec 7, 2026 – Dec 11, 2026	Cumulative final examination. Official date, time, location, and duration: to be confirmed.	Required core includes CLT, introductory inference, and Poisson processes.

Ten graded in-class activities are planned across Weeks 1–10, normally one selected activity per week. Other classroom practice and discussion work do not add graded submissions. Check Canvas for assessment instructions and the Weekly Guide for examination boundaries.

Assignments and grading

Assessment	Distribution and grading	Flexibility	%
Homework	7 sets, 60 points each. HW2 through HW7: 10 questions each (eight 5-point and two 10-point questions). HW1: six 10-point questions. All questions receive rubric grading.	Drop one lowest eligible set; retain 6 when all 7 are eligible.	25%
Weekly quizzes	10 Canvas quizzes; 5 two-point multiple-choice questions each (10 points).	Drop one lowest eligible quiz; retain 9 when all 10 are eligible.	5%
In-class activities	10 weekly activities in Weeks 1–10; 10 points each.	Drop one lowest eligible activity; retain 9 when all 10 are eligible.	5%
Midterm 1	In person; 55 minutes of working time.	No dropped examination score; approved arrangements handled separately.	20%
Midterm 2	In person; 55 minutes of working time.	No dropped examination score; approved arrangements handled separately.	20%
Final examination	Cumulative over the required core.	No dropped examination score. Official arrangements to be confirmed.	25%
Total	Discussion, Mathematics Review, and optional extra practice carry no grade.	Lecture attendance has no separate grade weight.	100%

How the dropped scores work

One lowest eligible score is dropped separately in homework, weekly quizzes, and in-class activities. Retained assessments within each of these categories count equally because they have equal point totals. Approved excusals and cancelled assessments are excluded before the drop. Unexcused missing work becomes zero after its final submission cutoff; an item awaiting grading is not a zero.

If only one eligible score remains in a category, it is retained. If none remain, contact the instructor to determine the appropriate arrangement. Approved accommodations are handled separately and are not exhausted by using a dropped score. Examination scores are not dropped.

Course percentage and letter grades

Your course percentage combines the category percentages using the weights in the table. Homework, quizzes, and in-class activities each exclude one lowest eligible score. Examination scores are not dropped. The letter-grade scale will be announced when confirmed.

Homework: submission, deadlines, and feedback

Homework must be handwritten and uploaded to Gradescope as one legible PDF, with pages mapped to questions. Paper scans and stylus handwriting are accepted. Typed solutions are not accepted unless required by an approved accommodation. Reopen the submitted PDF to check every page. Each subpart shows its points; state the relevant model, support, assumptions, method, and conclusion. Prefer exact expressions unless a decimal is requested, and round only the final answer.

All homework is due at 10:00 p.m. Pacific. A no-penalty 24-hour grace period ends at 10:00 p.m. the following day. That cutoff is the final submission window: there are no routine extensions or submissions after it. Approved accommodations are handled separately.

Set	Release at 3:00 p.m.	Due at 10:00 p.m.	Final cutoff at 10:00 p.m.
1	Sep 28, 2026	Oct 6, 2026	Oct 7, 2026
2	Oct 5, 2026	Oct 12, 2026	Oct 13, 2026
3	Oct 19, 2026	Oct 27, 2026	Oct 28, 2026
4	Oct 26, 2026	Nov 3, 2026	Nov 4, 2026
5	Nov 2, 2026	Nov 12, 2026	Nov 13, 2026
6	Nov 18, 2026	Nov 24, 2026	Nov 25, 2026
7	Nov 23, 2026	Dec 2, 2026	Dec 3, 2026

Grading rewards justified reasoning and valid alternative methods, not just the final answer. One carried arithmetic error is not repeatedly penalized. Worked explanations are released after all relevant submission windows close. If an accommodated window prevents release before an exam, equivalent practice with explanations is provided. For a grading question, identify the specific rubric item and explain your reasoning.

You may discuss homework ideas and methods, but write the final submission independently and name collaborators. Do not copy completed solutions or submit work you cannot explain. The academic-integrity and AI rules below apply.

Weekly quizzes

Quizzes are short individual checks with explanatory feedback. Complete quizzes individually in Canvas: one attempt and no time limit. Your own notes and course readings are permitted; consulting another person or using AI tools is not. All quizzes open at 3:00 p.m. and close at 10:00 p.m. Pacific on the dates below. There is no grace period or routine extension. Approved accommodations are handled separately.

Quiz	Opens at 3:00 p.m.	Closes at 10:00 p.m.
1	Oct 2, 2026	Oct 4, 2026
2	Oct 9, 2026	Oct 11, 2026
3	Oct 12, 2026	Oct 14, 2026
4	Oct 23, 2026	Oct 25, 2026
5	Oct 30, 2026	Nov 1, 2026
6	Nov 6, 2026	Nov 8, 2026
7	Nov 9, 2026	Nov 12, 2026
8	Nov 20, 2026	Nov 22, 2026
9	Nov 25, 2026	Dec 1, 2026
10	Dec 2, 2026	Dec 4, 2026

Full explanations are released after all applicable submission windows close. The early Quiz 3 window and the extended Quiz 9 window are intentional; follow the listed dates rather than assuming every quiz closes on Sunday.

In-class activities and absences

Begin on your own. When invited, compare your method with a nearby classmate. Then submit your own reasoning and one brief check or interpretation. The work must be completed in class; follow the collection instructions given that day. Uploads document work completed in class; they do not create a general take-home alternative. An ordinary missed activity is handled through the one-score drop in this category.

For serious illness, an emergency, or another exceptional absence affecting graded work, contact the instructor privately on Ed Discussion promptly about an equivalent activity or excusal. Approved disability and religious accommodations are considered separately from ordinary absence rules and dropped scores. TAs refer sensitive requests and decisions about exceptions to the instructor.

Examinations and recordings

Midterm 1 — Oct 16, 2026

During class, with 55 minutes of working time. Covers required material taught through Oct 9, 2026: events, counting, conditioning, independence, random variables, PMFs/CDFs, and Bernoulli/binomial laws. Expectation, variance, and joint laws are excluded, including the expectation lesson taught in the exam week.

Midterm 2 — Nov 16, 2026

During class, with 55 minutes of working time. Covers expectation, variance, discrete and continuous joint/conditional laws, covariance, elementary CDF transformations, Markov's and Chebyshev's inequalities, and the finite-variance weak LLN, through Nov 9, 2026. Earlier foundations remain available for use. The review on Nov 13, 2026 consolidates this scope. CLT, inference, and stochastic processes are excluded from Midterm 2.

Cumulative final

Covers the required core, including the CLT, introductory mean intervals with known population standard deviation, and Poisson counts and processes. All Markov-chain material and marked optional extensions are excluded; Markov's inequality remains required. The official date, time, location, and duration during Dec 7, 2026 – Dec 11, 2026 remain to be confirmed.

Permitted materials and examination conflicts

Updated October 15, 2026. For Midterm 1 on October 16, an instructor-provided formula sheet will be available to review on October 15 and supplied in print with the exam to all students, including those with DRC accommodations. Students may also bring one single-sided 8.5 × 11-inch page of their own notes. The provided sheet is additional and does not replace this personal sheet.

For Midterm 2 and the final, an instructor-provided formula sheet may or may not be supplied, at the instructor's discretion. Whether a sheet will be provided and what personal notes are permitted will be announced on Ed Discussion in advance, with time for students to prepare.

A basic nonprogrammable, nongraphing calculator is permitted; it is not required for Midterm 1. No other notes, communication, or electronic resources are permitted unless expressly authorized. Approved accommodations may change arrangements. Contact the instructor promptly about a conflict or emergency; TAs do not independently approve makeup exams.

Recordings

Recordings are not routinely distributed as substitutes for attendance. Where recordings exist, access may be provided for approved accommodations and instructor-approved serious illness, emergencies, or comparable circumstances. Approved recording accommodations are handled separately from this ordinary access policy. Availability will be confirmed before recordings are promised.

Academic integrity

Submit your own work, acknowledge collaborators and sources, and follow each assessment's permitted-resource rules. Do not copy, purchase, distribute, or obtain unauthorized solutions or examination content. Do not submit work from another course without prior permission. Suspected misconduct is handled under the UCSC academic-misconduct process.

AI tools may not be used to generate, solve, check, or rewrite assessed work, including homework, quizzes, in-class activities, and exams. Ask the instructor before using a resource if its status is unclear. You should be able to explain and reproduce your reasoning independently.

Student hours for this class

Plan for about 15 hours per week on this 5-unit course, including class meetings. The example below is a study plan, not an attendance or time-reporting requirement. Time spent on an optional discussion, tutoring, or office hours can replace part of the independent-study allocation. Needs vary by topic and exam week.

Activity	Example weekly allocation
Scheduled lectures	3 hours 15 minutes
Reading and reviewing definitions	3 hours
Homework and related problem practice	5 hours
Review, exam preparation, and help sessions	3 hours 15 minutes
Weekly quiz	30 minutes
Total	15 hours

Accessibility and disability accommodations

Your access to course materials and activities matters. If you need disability-related accommodations, contact the Disability Resource Center (DRC) and share your Accommodation Authorization Letter with the instructor through the appropriate private process. Please do this as early as possible so arrangements can be made; accommodation requests are not limited to the first two weeks. Contact the instructor promptly if a document, technology, or classroom activity creates an access barrier.

Approved accommodations are separate from the ordinary deadline, attendance, recording, and score-drop rules. You do not need to disclose a diagnosis on Ed Discussion or to classmates. DRC: drc@ucsc.edu · 831-459-2089. See the DRC and campus accessibility resources.

Religious accommodations

If a course requirement conflicts with a religious observance or practice, contact the instructor as early as possible to discuss a reasonable accommodation without penalty. Explain the affected requirement and the arrangement you are requesting. The instructor and student may seek guidance from the Dean of Students. An approved religious accommodation is handled separately from ordinary missed-work rules and is not replaced by a dropped score.

Principles of community

We will maintain a learning environment that respects differences in background, identity, and experience. Listen carefully, allow others to participate, ask questions without disparaging classmates, and challenge mathematical claims through definitions, assumptions, and evidence. Discrimination, harassment, and exclusion are inconsistent with this course's expectations and UCSC's Principles of Community. Respect others' privacy, while recognizing that university reporting obligations may apply.

Title IX, CARE, and student services

Title IX and confidential support

UCSC's Title IX Office addresses sex- and gender-based discrimination, including sexual harassment, sexual violence, and pregnancy-related discrimination. It can explain reporting options and supportive measures. Supportive measures may be available without a formal investigation. Before sharing sensitive information with course staff, ask about confidentiality and reporting responsibilities; the confidential services below provide a separate route for support.

For confidential support and advocacy, contact CARE (Center for Advocacy, Resources & Empowerment): care@ucsc.edu · 831-502-2273. For confidential counseling, contact CAPS: 831-459-2628. To contact the Title IX Office: 831-459-2462. For an immediate emergency, call 911. See Title IX information and reporting options and the UCSC Campus Resources directory.

Counseling and Psychological Services (CAPS)

CAPS offers confidential counseling and support for personal or psychological concerns affecting well-being or academic work. Contact CAPS at 831-459-2628; consult its website for current services and access information.

Learning Support Services and tutoring

Learning Support Services (LSS) offers academic support and tutoring. Check current offerings and enrollment information; CSE 107-specific tutoring has not yet been confirmed. Any confirmed tutoring arrangements will be announced on Ed Discussion. Course discussion sections remain optional and separate from LSS services.

Slug Support and basic needs

If food, housing, financial, or other urgent personal circumstances are interfering with your studies, Slug Support can help connect you with campus assistance. Contact the Dean of Students: deanofstudents@ucsc.edu · 831-459-4446. You may also tell the instructor privately that you need help, without sharing sensitive details publicly.

Technology and equipment

For account, Canvas, or other campus technology problems, contact ITS: help@ucsc.edu · 831-459-4357, or submit an ITS support ticket. Library laptop loans can help with short-term equipment access. Report a submission problem promptly and retain relevant error messages; approved arrangements are handled by the instructor.

Additional campus resources

The UCSC Campus Resources directory links accessibility, counseling, learning support, equipment loans, identity-based resource centers, and other services. Student Success provides further information about campus support programs.

Student feedback and course updates

You are welcome to share constructive feedback about pacing, clarity, workload, and accessibility during the quarter. At the end of the quarter, the Student Experience of Teaching survey provides an opportunity for anonymous feedback to help improve the course.

Canvas lists the schedule and gradebook; Gradescope homework windows will match the stated due dates and grace cutoffs. All deadlines use Pacific time, including the change from daylight to standard time. Changes will be announced and reflected in the course documents. The final-exam arrangements, letter-grade scale, and any course-specific tutoring remain to be confirmed.